

Assignment II

SDG-Based Questions – CO2

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1. Model Formulation

To predict whether a patient is **At Risk** ($y=1$) or **Not At Risk** ($y=0$), we define two features:

- x_1 : **Body Mass Index (BMI)**
- x_2 : **Daily Calorie Intake (in 100s of kcal)** (e.g., 1500 kcal $\rightarrow$ 15)

Activation & Output Rule

The weighted sum z and step activation function $f(z)$ are defined as:

$$z = w_0 + w_1 x_1 + w_2 x_2$$

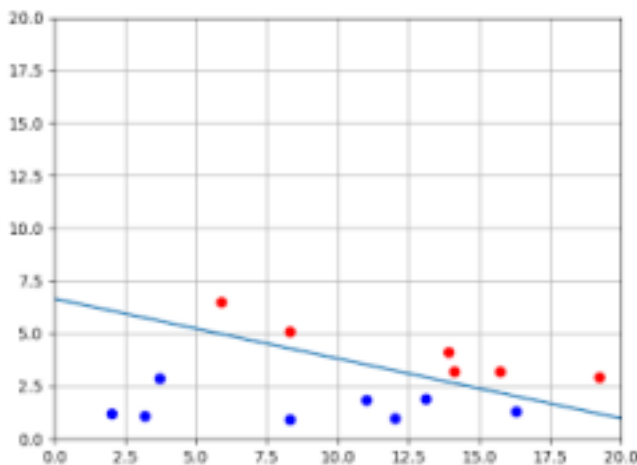
$$y^{\wedge} = f(z) = \begin{cases} 1 & \text{if } z \geq 0 \text{ (At Risk of Malnutrition)} \\ 0 & \text{if } z < 0 \text{ (Not At Risk)} \end{cases}$$

Here, w_0 is the bias (using a dummy feature $x_0=1$), and w_1, w_2 are the weights for BMI and Calorie Intake, respectively.

Weight Update Rule

Whenever the predicted class $y^{\wedge}$ differs from the target label y , we update weights using learning rate η :

$$w_i(\text{new}) = w_i(\text{old}) + \eta \cdot (y - y^{\wedge}) \cdot x_i$$



Perceptron Linear Decision Boundary. Source: Thomas Countz

2. Sample Dataset & Initial Hyperparameters

Let's use a learning rate of $\eta=0.1$ and initialize all weights to zero:

$$w(0)=[w_0, w_1, w_2]=[0.0, 0.0, 0.0]$$

Patient BMI (x1) Daily Intake in 100s (x2) Malnutrition Risk Label (y)

Patient A 16.5 12.0 (1200 kcal) **1** (At Risk) **Patient B** 22.0 22.0 (2200 kcal)
0 (Not At Risk)

3. Two Iterations of Weight Updates

Iteration 1: Patient A (x1=16.5, x2=12.0, y=1)

1. **Compute Weighted Sum (z):**

$$z=0.0(1)+0.0(16.5)+0.0(12.0)=0.0$$

2. **Predict Output ($y^{\wedge}$):** Since $z \geq 0$, $y^{\wedge}=f(0.0)=1$.

3. **Compare Prediction with Target:**

○ Target $y=1$, Prediction $y^{\wedge}=1$.

○ **Error:** $(y-y^{\wedge})=1-1=0$.

4. **Update Weights:** Since there is no error, weights remain unchanged:

$$w(1)=[0.0, 0.0, 0.0]$$

Iteration 2: Patient B (x1=22.0, x2=22.0, y=0)

1. **Compute Weighted Sum (z):**

$$z=0.0(1)+0.0(22.0)+0.0(22.0)=0.0$$

2. **Predict Output ($y^{\wedge}$):** Since $z \geq 0$, $y^{\wedge}=f(0.0)=1$.

3. **Compare Prediction with Target:**

○ Target $y=0$, Prediction $y^{\wedge}=1$.

○ **Error:** $(y-y^{\wedge})=0-1=-1$ (Misclassification).

4. **Update Weights ($\eta=0.1$):**

$$w_0(2)=0.0+0.1 \cdot (-1) \cdot 1=-0.1$$

$$w_1(2)=0.0+0.1 \cdot (-1) \cdot 22.0=-2.2$$

$$w_2(2)=0.0+0.1 \cdot (-1) \cdot 22.0=-2.2$$

$$w(2)=[-0.1, -2.2, -2.2]$$

Final Weights after Two Iterations

After two sample updates, the learned decision boundary equation becomes:

$$-0.1 - 2.2 \cdot \text{BMI} - 2.2 \cdot \text{Caloric Intake} \geq 0$$

Higher values of BMI and caloric intake now yield negative sums ($z < 0$), predicting **0 (Not At Risk)**, while lower values lean toward higher risk—aligning with clinical expectations for malnutrition evaluation.